

Final Project – Topic List and Presentation Instructions

I generated a topic list based on your responses as well as past submissions (previous years) and other topics from CASI, our electives, etc. If you listed some topics as already of interest, I tried to make sure you got at least one of those and then split the rest up to get decent coverage. For example, you might have gotten a topic I knew you saw in an elective to help give the class a sense of it. There are still topics without someone assigned so you can see them.

There's no requirement that folks end up with unique project topics. You also aren't tied to your assignments here and can brainstorm more to add to the list. The idea here is to give you as broad a base to think about project topics as possible.

For each of your assigned topics, you need to:

1. Create a Google slide in the class slide presentation (use the template). This has already been copied over to make 2 copies for each of you – just put your name in two slides and use those, and don't edit one someone else is using.
2. Fill in the slide with the requested information.
3. We'll go around and present these in class, informally on **Monday, Nov. 3rd**.

Note that the order slides are created is likely based on your work order, and I may re-arrange them to put related topics together, so presentation order will just end up being by slide as we go around in class. Again, this is informal, so you can just read your slides, but classmates might be interested in your level of enthusiasm (Did the topic seem cool? Did you find neat applications?).

Creating the slide deck together is a communal learning exercise. You'll be able to reference each other's slides when thinking about what you might want to study. A database of topic ideas!

The presentation is [linked from Google drive](#) and has been shared with each of you.

What do you have to fill in on the slides for your two topics?

Some information about your topic:

- Generally, what is it?
- Where is it applied/what is it for?
- Anything else potentially useful you'd want to share (such as R packages)
- Any major sub-topics of particular interest? Describe them briefly.
- At least 2 example applications (including citations)

You need to do at least one of the last two bullets for each topic, ideally, do both. For example, if there aren't major sub-topic areas, give at least 2 example applications. If there are major sub-topic areas, you can just describe them briefly, which should lead to example applications pretty easily if someone did a Google search. You only get one slide per topic though, which is why doing both isn't required if there's a lot of sub-topics you are trying to list, etc. I recommend using Google Scholar to find example applications and get their citations.

The template slide and demo slide should help demonstrate this.

If you want to tackle a third topic, or do a slide on a topic not in the list, please feel free to. Just copy over the second slide (template slide) and fill it in. The fourth slide is a list of “extra” pick-ups, so we don’t duplicate those. (And it’s ok if no one “picks up” anything extra – I might fill in some!)

Everyone please do not edit the second slide in the deck (template slide). If it ends up altered or you can’t find it (deleted), please let me know ASAP (I saved a back-up copy separately).

Other notes:

- These slides probably have more words on them than normally expected. That’s because we are creating this partially as a database of topic ideas. It’s ok to not read over the whole thing in class and it’s fine to have a lot of words (see the demo slide).
- You don’t have to read the names of all the example application papers out loud. You also don’t have to read those papers, but if you did glance at them or their abstracts and thought something was cool, your classmates would likely appreciate knowing. For applications, I expect you to say things like “It was pretty easy to find applications.” Or “I found applications in healthcare easily but not when I looked at ecology.” Or “I thought it was neat to see this application suggests it is spatial when this method isn’t usually applied in that realm” (demo slide example).
- Citations – to look up your topics you might be using the internet, your textbook (some of these are in there), other books, etc. The example articles you find might have enough info too, so you could even start there. However, there isn’t a ton of room on the slides to put this information (like an extra two textbook citations, etc.). What I’m envisioning we generate is a high-level overview, so that most sources would have the same basic information, and as such, just for this assignment, you don’t have to list where you got the info from on the slide (to save room). But you should have and keep this information in case someone asks about your slide(s). E.G. if someone really liked the topic you presented, they should be able to ask you where you got the info and you should be able to tell them / give them that info.
- Finally, you’ve probably seen in the list that I tried to list “categories” of the topics. There’s no one “right” categorization of these topics. Example: Principal components analysis (PCA) is both a multivariate data analysis topic and a topic in dimension reduction. It’s a linear method too. It’s also used as pre-processing in machine learning because it can be used for dimension reduction. Where does it go? Any of them. I tried to provide one option for category but you should know most of these fit in multiple places or have variants for different categories. For example, there is a functional data analysis version of PCA called functional PCA (FPCA). If something doesn’t have a category in the list, it’s because I thought it was so general as to be the category or didn’t have a category pop into my head.